

ThaiEditBench: A Mechanism-Typed, Length-Stratified Benchmark for Thai Orthographic Error Correction

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Abstract

Thai is written without word boundaries and layers tone marks, silenced consonants, and heavy loanword transliteration onto a Brahmic script. This produces orthographic error mechanisms unlike those targeted by English grammatical error correction (GEC) or Chinese spelling check (CSC). We present **ThaiEditBench**, a character-level Thai orthographic error-correction benchmark with three components: (i) a principled **six-class mechanism taxonomy** plus an etymology tag; (ii) **three length tiers** (sentence, paragraph, and page) that isolate robustness to long, mostly-clean context; and (iii) **contamination-robust construction**, in which errors are *injected* into novel clean text and *mined* from Wikipedia cleanup-bot history, then screened by a memorization probe. Because errors are introduced by construction, gold corrections are exact and require no human annotation. Grading is at the character-cluster level with span-based matching adapted from ERRANT. Evaluating **16 models**, we find that (1) on this benchmark, correction is primarily a *detection* problem rather than a knowledge problem; (2) the evaluated Thai-specialized models do not lead the strongest general models; and (3) long, mostly-correct context exposes **over-edit accumulation** and clean-text fidelity failures that sentence-level evaluation hides.

1 Introduction

Thai orthography is error-prone in script-specific ways: tone-mark selection and placement, silenced-consonant marks, homophonous consonants, vowel-form confusions, consonant-cluster metathesis, and inconsistent loanword transliteration. These are **character-level** errors that turn on **specific mechanisms**, and Thai’s lack of whitespace word boundaries makes them hard to localize and score with token-based tools. Yet the resources to study them are missing. Grammatical-error-correction benchmarks are English-centric (Bryant et al.,

2017, 2019; Napoles et al., 2017), and the closest character-level analogue is Chinese spelling check (Tseng et al., 2015; Liu et al., 2011). For Thai, existing error-annotated resources are social-media, informal, or learner-domain (Lertpiya et al., 2020; Nakwijit and Purver, 2022), and there is no formal-register, mechanism-typed, contamination-controlled orthographic benchmark. Document-level and context-aware GEC study correction over long context (Yuan and Bryant, 2021), but no benchmark isolates length and error-sparsity robustness for Thai orthographic correction.

Contributions. (1) **ThaiEditBench**, a character-level Thai orthographic correction benchmark: a six-class mechanism taxonomy plus an etymology tag, **three length tiers** (sentence, paragraph, page), built **contamination-robust by construction** (confusion-set injection over novel clean text and Wikipedia cleanup-bot mining, memorization-probe screened) so gold is exact and needs no human annotation. (2) An **ERRANT-style orthographic edit-typer for Thai** that aligns edits at TCC level and labels each with one of the six mechanism classes, enabling **per-mechanism span-F0.5** (cf. ChERRANT’s dual-granularity scoring (Zhang et al., 2022)). We use it as a scoring and diagnosis instrument, not a general Thai grammatical error classifier. (3) A **16-model evaluation** yielding three findings: correction-is-detection, the evaluated Thai-specialized systems do not lead, and over-edit accumulation under long context in weaker models. We release code, data, grader, edit-typer, and the variant map, anonymized for review: <https://github.com/BlindReviewYN/ThaiEditingBench>.

2 Related Work

GEC and CSC. English grammatical error correction is evaluated with span-based metrics. The M^2 scorer matches a system’s edits against gold edit

spans (Dahlmeier and Ng, 2012), ERRANT (the ERROR ANnotation Toolkit) additionally infers an error type for each edit automatically and reports an F0.5 score (Bryant et al., 2017), and GLEU measures fluency (Napoles et al., 2015). ERRANT-style span matching aligns the wrong and corrected text into a minimal set of edit spans, then credits a system for each gold span it recovers with the correct replacement; we adapt this scheme to Thai TCC units (Section 4.2). Character-level Chinese spelling check is the closest task, scoring position-based detection and correction F1 without word boundaries (Tseng et al., 2015; Wu et al., 2013; Yu et al., 2014), with confusion-set construction (Liu et al., 2011) and the MuCGEC/ChERRANT dual-granularity scorer (Zhang et al., 2022). ERRANT has been ported to Arabic (Belkebir and Habash, 2021) and Korean (Yoon et al., 2023), but not Thai. Wikipedia-revision mining of typo pairs is established for English (Grundkiewicz and Junczys-Dowmunt, 2014) and Japanese (Tanaka et al., 2020).

Thai NLP. We build on PyThaiNLP for tokenization and dictionary tooling (Phatthiyaphaibun et al., 2023) and on the Thai Character Cluster as the minimal orthographic unit (Theeramunkong et al., 2000). Prior Thai spelling correction targets social text (Lertpiya et al., 2020, 2023), and related work studies the semantics of Thai misspellings (Nakwiji and Purver, 2022). The VISTEC-TP-TH-2021 Twitter corpus includes misspelling detection and correction as well as word-segmentation annotation, but it is informal Twitter text rather than a formal-register, mechanism-typed orthographic benchmark (Limkonchotiwat et al., 2021).

3 The ThaiEditBench Benchmark

3.1 Task

Given a Thai passage containing character-level orthographic errors, return the passage with those errors corrected and **nothing else changed**: no rewriting, re-spacing, numeral, or punctuation edits. We score under a strict **minimal-edit contract**, where any change to a non-erroneous span is an *over-edit* and is penalized.

3.2 Mechanism taxonomy

We define six orthographic mechanism classes, each carrying an **etymology tag**: tone mark, consonant-pair, silent-consonant, vowel, cluster or metathesis, and loanword transliteration. Worked

examples per class are in Appendix A. Canonical forms are verified against the Royal Institute Dictionary (ORST). Class 6 is governed by a **loanword-absorption rule**: only *contemporary* foreign loans are class 6, while long-integrated borrowings (Tamil, Khmer, Persian, and Sanskrit–Pali doublets) are treated as native.

3.3 Construction

Two complementary sources both yield **construction-gold**. Applying the gold edits to an item restores the clean original, which we verified holds for 100% of items.

- **(a) Confusion-set injection.** A 1,044-pair $\times$ 6-class Thai confusion set is injected into *novel clean text* using **whole-token-run matching** (not substring) to protect short valid forms. Each injected item records the Thai Character Cluster (TCC) span, class, and etymology.
- **(b) Cleanup-bot mining.** Real wrong $\rightarrow$ correct pairs are mined from the edit history of a Thai Wikipedia spelling-cleanup bot (RIDgated and misalignment-guarded), following the revision-mining tradition (Tanaka et al., 2020; Grundkiewicz and Junczys-Dowmunt, 2014).

Registers. The clean text spans news (thaisum), encyclopedic (Thai Wikipedia), and government and legal (Royal Gazette, thaigov), all permissively redistributable.

3.4 Contamination control

The benchmark reduces contamination risk by evaluating systems on **corrupted passages that do not appear verbatim in the source corpora**. A residual risk remains: a model that memorized a clean source passage could reconstruct it directly instead of detecting and correcting the injected errors. We therefore run a **verbatim-continuation memorization probe** on the source passages and hold the lowest-memorization material out from the public release.

3.5 Length tiers

The same pipeline produces three tiers that vary passage length and error sparsity. **T1** (sentence) has 795 items of ~ 100 characters each; **T2** (paragraph) has 223 items of ~ 424 characters; and **T3** (page) has 124 items of $\sim 1,242$ characters. Item counts are class-balanced across the six mechanisms, and

full per-tier statistics are in Appendix C. The error budget is **dense at the sentence level**, with roughly one error per 100 characters, and deliberately **sparse at paragraph and page length**, where most sentences are clean distractors. T2 and T3 therefore measure **precision under long correct context** (over-editing), not just recall, an axis the CSC and GEC literature does not test.

3.6 Quality control (fully automatic)

Because errors are introduced by construction, the gold is exact. We add two automatic guards.

- **Cross-model consensus audit.** When at least $k=3$ of the 16 systems produce the *same* normalized output that differs from gold, we flag the item for review. Consensus is a flagging heuristic, not proof of a gold defect. We drop the item only when the agreed output is a residual *source* typo that the construction-time filter missed. We keep the other two cases: a *shared miss*, where the consensus equals the corrupted input and the gold is fine, and a *systematic over-edit* on punctuation or register. The classification is deterministic. The audit removed 24 of 819 (T1), 27 of 250 (T2), and 26 of 150 (T3) items, and post-audit no item is universally missed. The full protocol is in Appendix B.
- **Variant reconciliation.** Spellings with two accepted forms are reconciled to the gold form before scoring, token-aware and compound-safe. This covers orthographic doublets, loanword transliterations, and an informal-to-formal pronoun pair, so a model is never penalized for a legitimate variant.

Orthographic correction is essentially **single-reference**, since there is one correct spelling, so we do not adopt multi-reference GEC machinery. Variant reconciliation covers the few genuine alternatives.

4 Evaluation Protocol

4.1 Metrics

We grade at the TCC level (Theeramunkong et al., 2000) with ERRANT-style span matching (Bryant et al., 2017). The **primary metric is correction F1**: minimal TCC edits are derived by aligning the wrong text against *both* the gold and the system output under the **same** Levenshtein alignment,

so an output identical to the reference scores exactly 1.0 (SIGHAN-CSC-comparable (Tseng et al., 2015)), and we report it with a 95% **item-bootstrap confidence interval**. **Detection F1** credits the correct error site and ignores the replacement. **Clean-sentence fidelity** is the fraction of error-free sentences left untouched, the exact and construction-grounded precision signal for T2 and T3. We also report **per-mechanism span-F0.5** via the edit-typer (Section 4.2). Word-level GLEU is reported only as a supplementary metric in Appendix D, because Thai word segmentation is unreliable for this task.

4.2 ERRANT-style edit-typer

We adapt the ERRANT/ChERRANT design (Bryant et al., 2017; Zhang et al., 2022) to Thai orthography. For a (wrong, correct) pair: extract edit spans at **TCC level** (the same alignment the grader uses), expand each to its enclosing word (PyThaiNLP newmm (Phatthiyaphaibun et al., 2023)), and label it with one of the six mechanism classes (plus an edit-operation tag); class 6 is etymology-gated by a loan lexicon. This enables **per-mechanism span-F0.5**. Its inventory is orthographic-mechanism-specific and reuses the construction-time classifier, so we treat it as a **scoring instrument**, not a general Thai grammatical error classifier.

4.3 Models and inference

We evaluate 16 models, each in a single fixed configuration, at temperature 0 where supported, with one fixed Thai instruction prompt. The **frontier proprietary** models are GPT-5.5 (medium and high reasoning), Gemini 3.5 Flash, Claude Opus 4.7 (non-thinking), Claude Sonnet 4.6 (non-thinking), and GPT-5.4-mini. The **open-weight** models are Gemma-4-31B, DeepSeek V4 Pro and Flash, GLM-5.1, MiniMax-M2.7, and Qwen3.6-35B-A3B. The **Thai-native** models are Typhoon 2.5 (30B) and three ThaiLLM 8B-instruct models served through the national ThaiLLM gateway, namely OpenThaiGPT, Typhoon-S, and THaLLE (Pipatanakul et al., 2024; Pipatanakul and Taveekitworachai, 2026).

5 Results and Analysis

5.1 Leaderboard

The full leaderboard is Table 3 (Appendix D), reporting correction F1 across the three tiers (post-audit: $n = 795/223/124$). The frontier clusters at

model	fixed	wrong_corr	missed
GPT-5.5 (high)	0.993	0.000	0.007
Gemma-4-31B	0.956	0.004	0.040
Qwen3.6-35B-A3B	0.817	0.018	0.165
Typhoon 2.5 30B	0.764	0.036	0.201

Table 1: Gold-edit outcome decomposition (T1). `wrong_corr` stays near zero for every model; the budget is missed.

0.95 to 0.97 on T1, and bootstrap CIs separate three strata (frontier, then GLM, then the Thai-8B models and runaways). No single model solves the entire T1 set, and no item is *missed* by every model, so the gold is clean post-audit yet retains per-model hard cases. **The benchmark discriminates**, and the length-degradation slope (T1 to T3) is the sharpest discriminator (Section 5.3).

General frontier models dominate, while Thai-specialized models trail. Typhoon 2.5 scores 0.742 on T1 and 0.467 on T3, and the 8B national-gateway models score 0.74 to 0.78 on T1 but fall to 0.45 to 0.62 on T3. The strongest **open-weight** model is a general 31B model, Gemma-4. It is within CI of the frontier proprietary models on T1 and T2 and is the single best model on the page tier (T3, 0.889). Under this setup the evaluated Thai-specialized systems do not lead, and the gap widens with length. This comparison is confounded by differences in model scale, serving stack, and inference configuration, so it is not evidence about Thai-specialized pretraining in general.

5.2 Correction is a *detection* problem

Decompose every gold edit (T1) into **fixed**, **wrong_corr** (right site, wrong fix), and **missed** (Table 1). Across **all 16 models**, `wrong_corr` never exceeds 0.036, while `missed` carries essentially the entire error budget. Once a model *notices* an error it almost always fixes it correctly. What separates models is **error discrimination**, not knowledge of correct spellings. This reframes the implicit assumption in correction-centric CSC work.

5.3 Length robustness

Sentence-tier rank does **not** predict paragraph or page behaviour, and the **degradation slope from T1 to T3 is the sharpest discriminator** (Figure 1a). The frontier loses only 0.10 to 0.12 correction F1 (GPT-5.5-med 0.969→0.856; Gemma-4 0.950→0.889), while weak models fall off a cliff (Typhoon 2.5 0.742→0.467; Qwen 0.705→**0.308**,

a drop of 0.40). The long tiers also expose over-editing. We measure it as the over-edit rate per 10,000 clean TCCs, which factors out how much clean text a model is exposed to. This rate is roughly flat or falls across tiers (Qwen 105→107→65; Typhoon 2.5 57→33→27; Figure 1b), so over-editing accumulates with length through exposure rather than through a rising false-positive rate. The discriminating signal is the *level* of that rate, not its slope: weak models over-edit at 5 to 20 times the frontier’s per-unit rate, which stays below ~15 per 10,000 clean TCCs at every tier. Clean-sentence **fidelity** tells the same story. The frontier holds at or above 0.95 throughout, while weak models sit at 0.78 to 0.90 regardless of length. **Over-editing and missing are two faces of the same weak discrimination**: a model that cannot tell correct from incorrect both misses real errors and rewrites correct text. The exception is instructive. **Gemini 3.5 Flash keeps high recall at T3, fixing 0.985 of gold edits, yet drops to 0.819 correction F1 purely on precision**. It is the one frontier model whose page-tier score is limited by over-editing rather than by detection.

5.4 Per-mechanism difficulty

Class 6 (loanword) is the hardest mechanism, with ~0.81 of edits fixed versus ~0.86 to 0.90 for tone marks, consonant pairs, and silent-consonant marks. The failure is again **missed**, not miscorrected: models are least certain that a transliteration is wrong. But the *ranking* within class 6 does **not** favour Thai-native models. Difficulty is a property of the **mechanism**, not of cross-lingual knowledge. Register agrees: loanword-dense encyclopedic text is hardest (recall ~0.71 to 0.82) versus government and news text (~0.83 to 0.90).

6 Conclusion

ThaiEditBench is a mechanism-typed, length-stratified, contamination-robust benchmark for Thai orthographic correction, released with an ERRANT-style edit-typer for per-mechanism scoring. Across 16 models it reframes correction as a **detection** problem, shows that the evaluated **Thai-specialized systems do not lead**, and surfaces **over-edit accumulation** under long context in weaker models.

Limitations

Sparse error density (primary). The long tiers (T2 and T3) are deliberately sparse, with 1 to 3 errors scattered across otherwise-clean passages, to isolate precision under long correct context. We therefore do **not** test **high error-density** regimes such as heavy-typo input, degraded OCR, or L2 writing, where errors co-occur, interact, mask one another, and correction order matters. Extending ThaiEditBench to dense multi-error passages, and testing whether the detection-not-knowledge result survives when errors crowd each other, is the most important next step.

Reasoning-mode not swept. Models run in mixed default configurations, so the leaderboard compares default behaviour rather than a controlled thinking-on or thinking-off contrast.

Configured-system comparison. Each system is evaluated under one fixed configuration. Reasoning modes, token budgets, provider defaults, serving stacks, and model scale are not controlled, so the leaderboard reflects configured-system behaviour rather than an intrinsic ranking of base-model capability.

Length-normalized over-editing. Raw over-edits per item rise with passage length, because longer passages expose more clean text where a false positive can occur. We therefore report the over-edit rate per clean TCC alongside the absolute per-passage count, and we base rate claims on the normalized metric while reading the absolute count as user-visible accumulation.

Minimal-edit vs fluency boundary. Occasionally a frontier “over-edit” is actually catching a residual error that our injection-gold did not mark, which slightly understates the best models.

No human inter-annotator agreement. Gold is construction-based, either injected or mined, and audited automatically. We run no human annotation pass, so no inter-annotator agreement is reported. This is sound for *injected* errors, where the gold is exact by construction, but the mechanism-class labels and the mined items rest on automatic typing rather than on measured human consensus.

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class 6. Long-integrated borrowings are treated as **native** even though etymologically foreign: Portuguese (กะลະແມ), Burmese (กะปီ), Mon (กะพง), Khmer (กบฏ), and Tamil, Persian, and Sanskrit–Pali doublets. Borderline cases (e.g. กังสุล, *consul*) are resolved by ORST listing + Wiktionary borrowed-term categories, and the tag is recorded per item so the rule is auditable.

B Gold quality control and the consensus audit

Why construction-gold still needs a cleanup pass. Injection guarantees that applying the gold edits recovers the clean original (verified 100%), so the *injected* error’s correction is exact. It does **not** guarantee that the **source passage was typo-free**: the permissive corpora contain a small rate of residual orthographic errors. When an item’s source carries such a residual typo, the gold inherits it, so a model that fixes *that* genuinely wrong span is unfairly scored as a miss or an over-edit. We remove these automatically.

Cross-model consensus audit. For every item we pool all 16 outputs (variant-reconciled) and look for a *consensus* output, the same normalized string from at least $k=3$ systems, that differs from gold. The discrimination is three-way. (i) If the consensus equals the corrupted input, this is a *shared miss*, the gold is correct, and we **keep** the item. (ii) If the consensus is a punctuation, whitespace, or register variant, this is a systematic *over-edit*, the precision signal we want, and we **keep** it. (iii) If the consensus is a *different* spelling fix at an unmarked span, this is a residual *source* typo, and we **drop** it. Only case (iii) is a gold defect, and the explicit test of whether the consensus equals the input prevents conflating it with shared misses and over-pruning. The audit removed 24 (T1), 27 (T2), and 26 (T3) items, for example ทั้งสิน→ทั้งสิน, ปฐิเศษ→ปฐิเสส, and โดบ→โดย, all residual *source* typos. **Post-audit invariant:** no item in any tier is missed by all 16 models. This is the signature that residual gold defects are gone while genuinely hard items remain. The drop list is committed by the deterministic three-way rule above. The audit also ships with a self-contained HTML review view (colored gold-vs-consensus diffs, with DROP, KEEP, and VARIANT badges) for optional inspection and reproducibility.

tier	unit	items	chars	err/it	news/gov/enc
T1 sent.	1 sent.	795	100	1.23	275/368/152
T2 para.	≈5 sent.	223	424	1.24	80/79/64
T3 page	≈15 sent.	124	1,242	1.56	42/43/39

Table 2: Per-tier statistics (post-audit; Section 3.6). Counts are class-balanced across the six mechanisms (T1: 115 to 211 edits per class). The described tiers are 100% confusion-set injection; cleanup-bot material is held out from the public release.

C Construction details

Confusion set: 1,044 canonical pairs $\times$ 6 classes, RID-verified against `thai_orst_words` (87.5%; misses are modern loanwords absent from the offline ORST), each tagged with etymology and edit-operation. Injection uses **whole-token-run matching** so the 51 valid short forms (≤ 3 characters) that coincide with a confusion target are never corrupted. Cleanup-bot mining draws pairs from the `usercontribs` history of a single Thai Wikipedia spelling-cleanup bot, RID-gated and misalignment-guarded; these are held out from the public release. A verbatim-continuation **memorization probe** (Typhoon 2.5 30B) scores all sources LOW (mean longest-common-prefix 0.003 to 0.017), and the lowest-memorization text is held out from the public release.

Release and license. We release the benchmark for research use under CC BY-SA (data) and the MIT license (code), consistent with the permissive redistribution terms of the source corpora; the anonymized review release is linked in the contributions (Section 1).

D Full results

Pooled per-class outcome split (T1, all 16 models).

class	gold edits	fixed	wrong_corr	missed
1 tone	2,608	0.896	0.011	0.092
2 consonant	2,192	0.876	0.006	0.117
3 silent-consonant	2,640	0.887	0.010	0.103
4 vowel	2,896	0.857	0.002	0.140
5 cluster	3,392	0.863	0.012	0.126
6 loanword	1,968	0.808	0.003	0.189

Over-edits per 100 items across tiers (absolute accumulation).

model	T1 det	T1 corr	T2 corr	T2 fid	T3 corr	T3 fid
GPT-5.5 (high)	0.973	0.973	0.969	0.990	0.855	0.974
GPT-5.5 (med)	0.969	0.969	0.955	0.983	0.856	0.971
Claude Opus 4.7	0.971	0.968	0.948	0.986	0.848	0.974
Gemini 3.5 Flash	0.969	0.965	0.960	0.983	0.819	0.960
Gemma-4-31B	0.954	0.950	0.954	0.988	0.889	0.981
GPT-5.4-mini	0.959	0.949	0.917	0.986	0.853	0.988
Claude Sonnet 4.6	0.952	0.947	0.906	0.977	0.824	0.972
DeepSeek V4 Flash	0.952	0.947	0.945	0.986	0.828	0.966
DeepSeek V4 Pro	0.928	0.926	0.926	0.970	0.746	0.954
GLM-5.1	0.851	0.846	0.812	0.961	0.665	0.954
Typhoon-S 8B [†]	0.788	0.782	0.702	0.961	0.619	0.965
THaLLE 8B [†]	0.758	0.749	0.595	0.775	0.517	0.829
Typhoon 2.5 30B [†]	0.776	0.742	0.631	0.881	0.467	0.896
OpenThaiGPT v7.2 [†]	0.755	0.741	0.477	0.887	0.454	0.896
MiniMax-M2.7	0.747	0.740	0.720	0.974	0.470	0.968
Qwen3.6-35B-A3B	0.721	0.705	0.383	0.776	0.308	0.836

Table 3: Leaderboard: correction and detection F1 and clean-sentence fidelity across the three length tiers. **Bold** = column best; [†] = Thai-native. Per-model CIs, detection F1, ChERRANT span-F0.5, and per-class recall are released with the data. The frontier T1 CI width is $\sim \pm 0.01$. **Gemma-4-31B leads T3 (0.889)**, clear of the GPT-5.5 pair near 0.855, so a *general* open-weight model is the most robust at length.

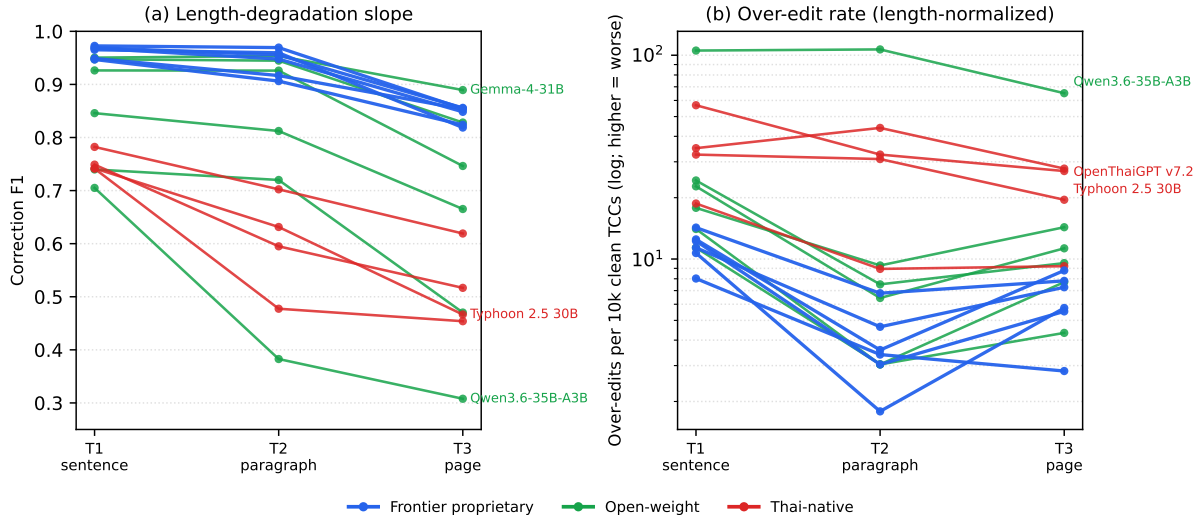


Figure 1: Length robustness. (a) Correction-F1 slope from T1 to T3. The frontier (blue) and the strongest open-weight model (Gemma-4, green) stay high while weak models slope off. (b) Over-edit rate per 10,000 clean TCCs on a log scale, where higher is worse. The rate is roughly flat across length, so the per-passage accumulation (the absolute counts below) reflects text exposure rather than a rising false-positive rate. Weak models over-edit at many times the frontier’s per-unit rate at every tier.

model	T1	T2	T3
GPT-5.5 (med)	7	8	41
Gemma-4-31B	6	8	32
GLM-5.1	10	23	106
Typhoon 2.5 30B	32	82	201
OpenThaiGPT v7.2	20	110	206
Qwen3.6-35B-A3B	59	268	484

model	T1	T2	T3
GPT-5.5 (med)	12.3	3.0	5.5
Gemma-4-31B	11.4	3.0	4.3
GLM-5.1	17.8	9.3	14.3
Typhoon 2.5 30B	56.8	32.6	27.0
OpenThaiGPT v7.2	35.0	44.0	27.8
Qwen3.6-35B-A3B	105.4	106.8	65.1

Over-edit rate per 10,000 clean TCCs (length-normalized; the Figure 1b data). The per-unit rate is roughly flat or falls across tiers, so the per-item growth above reflects text exposure, not a rising false-positive rate.

Register effect (pooled correction recall, T1/T2/T3). encyclopedic 0.82/0.79/0.71; news 0.87/0.85/0.76; government 0.89/0.90/0.83. Per-model $\times$ class correction recall and ChERRANT span-F0.5 per type are released with the data.

Word-level GLEU is released as a supplementary comparability metric only. We exclude it from the main metrics because Thai word segmentation is unreliable for this task.

E Prompt and inference

A single fixed Thai instruction prompt is used for all models. It asks the model to return the passage with orthographic errors corrected and **nothing else changed**, explicitly forbidding spacing, punctuation, and numeral edits. Temperature is 0 where supported. Claude models use the Anthropic Message Batches API in non-thinking mode. GPT-5.x models use the Responses API at the stated reasoning effort with a 3k-token budget. Open reasoning models run in native thinking mode with large token budgets, and token-runout counts as a failure. ThaiLLM-gateway models are reached with a browser User-Agent header, because the gateway WAF otherwise blocks the default SDK agent.